Size, Shape and Flow Characterization of Ground Wood Chip and Ground Wood Pellet Particles

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Abstract

Size, shape and density of biomass particles influence their transportation, fluidization, rates of drying and thermal decomposition. Pelleting wood particles increases the particle density and reduces the variability of physical properties among biomass particles. In this study, pine chips prepared for pulping and commercially produced pine pellets were ground in a hammer mill using grinder screens of 3.2, 6.3, 12.7 and 25.4 mm perforations. Grinding pellets produced the smaller particles with narrower size distribution than grinding chips. Wood chips consumed about 7 times more specific grinding energy than to pellets to produce same size of particles. Shape factors measured by digital image analysis showed that chip particles were had the aspect ratios 3 times larger than pellet particles. Pellet particles were more circular shape. The mechanical sieving underestimated the actual particles' size and did not represent the size of particles correctly. Instead, digital imaging is preferred. Angle of repose and compressibility tests showed pellet particles made a less compacted bulk, had lower cohesively and did flow easier in a pile of particles. A quantified particle shape factor influenced flow properties more than the particle dimensions did?.

Keywords: Size, Shape, Flowability, Shape factor, Ground chip, Ground pellet.

1. Introduction

Power generation plants use woody biomass in the forms of either wood chips or wood pellets [1-3]. Wood pellets are denser in bulk and more homogenous in physical properties than wood chips [4, 5]. Wood pellets provide a convenient form of biomass for its handling, storage and feeding to the pyrolysis/combustion reactors [3, 6]. Due to the heat and mass transfer limitations of using the large chips or pellets in the reactors, it is recommended to grind the

biomass to particles smaller than 2 mm to be fed to the thermochemical processes, fast pyrolysis reactors and co-firing boilers [7-12]. Size, shape and density of the ground biomass particles may affect the reactor feeding system, flow properties, ignition temperature and kinetics of drying and thermal decomposition [4, 12-14].

Accurate, fast and cheap particle size analysis is desirable in the industries that handle particulate materials. One of the conventional methods to obtain the particle size distribution is mechanical sieving. American Society of Agricultural and Biological Engineers (ASABE Standard S424.1, 2007) developed the mechanical sieving method as a standard particle size analysis for biomass particles. Mechanical sieving determines the mass percent of particles retaining on each sieve [15]. Particles pass through the sieves based on their width. Length of the particles is ignored in a sieving process. Biomass particles are mostly irregular and heterogeneous in size and shape [16]. Then, two biomass particles that pass through the same sieve may have different shapes. That is why the information obtained from the sieving process may not represent the size and shape characteristics of irregular-shaped biomass particles. Conducting the mechanical sieving for numbers of samples is a laborious work [17], and may not be consistent [18].

Rather than the conventional mechanical sieving approach, the advanced techniques such as machine vision are able to analyze the particle size and shape using the image analysis techniques [19]. Image analysis is a practical method to determine the actual dimensions and shape of single particles. Image analysis is not subjective and is repeatable over the same picture [15, 20]. The bottleneck of image analysis is two dimensional analysis that misses thickness of particles [18, 20]. However, this shortage returns to the difficulty in picturing the thickness of an irregular-shape particle. Tannous et al. [16] reported that picturing and measuring the thickness

of a single Douglas fir particle is not practical and reliable and assumed that particles' thickness is 30% of their width. A simple and open source software to conduct the image analysis is ImageJ. ImageJ is developed by the National Institute of Health (NIH) of USA. It is a Java based image processing software that can be freely downloaded.

To describe the particle shape, literature established a series of shape factors to evaluate the form and structure of the particles. Shape factors are quantitative information about particle shape. Riley [21] stated the difficulty of 3D picturing of particles and introduced projected shape factors that were based on width, length, inscribed and circumscribed circles. Trottier et al. [22] defined shape factors using two or three dimensions of an individual particle and categorized shape factors in four groups: (1) dimensional ratios, (2) sphericity that evaluate the deviation of a particle from a sphere/circle, (3) roundness and circularity that show angularity and sharpness of corners, and (4) roughness that shows surface structure. Figure 1 shows the concept of various shape factors in a single wood particle.

The comprehensive information on the actual size, shape and density of biomass particles is crucially important in all applications that the particles need to flow. Hoppers, screws, pneumatic transportation in pipelines and feeding systems in thermochemical conversion processes are the most applicable examples. Physical properties such as particle size, particles shape and surface roughness affect the flowability of biomass particles [16, 23-27]. Biomass particles are cohesive and may stick together in a flowing stream of particles and cause a variety of flow issues. Bridging of biomass particles in feeding systems of thermochemical conversions is a common industrial issue [28, 29]. The convenient analyses to evaluate the flow properties of minerals are mostly used to conduct the flow characterization of biomass particles [16, 25, 26, 30-37]. Loose and tapped bulk densities indicate the first and quickest estimation of particles'

flowability [32, 33]. In tapping process, particles re-arrange and change the packing condition [32]. The more difference of bulk density before and after the tapping process shows the particles' more tendency to make a compact bulk. Two dimensionless numbers of Hausner ratio (HR) and Carr compressibility index (CCI) quantify the increase in bulk density during tapping process [16]. Open literature shows that size and shape of particles, the way the particles arrange in the bulk and friction among the particles change the bulk density [32, 33, 38]. Wu et al. [23] measured the bulk density of wood chips and wood pellets and showed that the bulk density of pellets is about 2-3 times of wood chips. Wu et al. [23] did not compare the bulk density of ground chip and ground pellet particles. Lam et al. [33] measured the bulk density of wheat straw and switchgrass as a function of physical attributes. Tapping the bulk compressed the longer particles more than shorter ones. The exact HR and CCI values strongly depend on the technique to determine the tapped bulk density, dropping height and number of taps. Tannous et al. [16] showed that more tapping increases the tapped bulk density and then HR value. That is why the published values of tapped bulk density are seldom comparable. Although, implementing a consistent procedure of bulk density measurement is necessary to study the relative compressibility of different samples. In a comprehensive fluidization review, Geldart et al. [39] showed that particles with HR of less than 1.25 are free-flowable and easy to fluidize. Particles with HR of higher than 1.4 are cohesive and difficult to fluidize. Particles with HR values of 1.25 to 1.4 are in the middle and have partial properties of both groups. The CCI values between 5 and 15, 12 and 16, 18 and 21, and 23 and 28% indicated excellent, good, fair, and poor flowability, respectively [16]. Tannous et al. [16] showed that HR and CCI values for Douglas fir particles decreased with particle size. By decreasing the particle size in Sauter diameter range of 74μm-781μm, HR and CCI values decreased from 1.85 to 1.32 and from 45.84 to 24.06,

respectively. Tannous et al. [16] showed that smaller particles packed more with tapping and have poor flow properties that larger particles.

Angle of repose (AOR) is another test to study the relative flowability of samples. AOR is the angle of piled particles respect to the horizontal surface. AOR indicates the failure properties of particles under gravity [27]. AOR depends on cohesiveness and stickiness of particles. Literature discussed that increase in particle size is accompanied by a decrease in cohesiveness [16, 30]. It is suggested that the angle of repose below 30° shows good flowability, 30°-45° some cohesiveness, 45°-55° true cohesiveness and higher than 55° very high cohesiveness [30]. In some other studies [31, 35, 40], 40° was mentioned as the criteria of free flowability. Reviewing the literature shows that there is no general agreement on the test procedure and the scale of AOR equipment. Geldart et al. [30] showed that variable amount of material may change AOR values. Wu et al. [23] compared AOR of wood chips (lengths of 20, 40 and 100 mm) and wood pellet (diameters of 6, 8 and 12 mm). Wu et al. [23] showed that wood pellets with AOR values of 35-38° were more flowable than wood chips with AOR values of 44-46°. Wu et al. [23] did not measure the AOR of ground chip and ground pellet particles.

Objectives

Size, shape and density of the biomass particles affect the flow properties of the biomass particles. Reviewing the open literature shows that there is limited information on physical characterization of ground woody biomass particles in two forms of ground chip and ground pellet regarding their actual dimensions, shape, density and flow properties. Studies on biomass density has been attributed to the type of testing biomass and the effect of density of same biomass species is not well documented in the literature.

The objectives of this research are to study the size, shape, density and relative flow properties of ground pine chip and ground pine pellet particles. Digital image analysis determines the dimensions, shape and some pre-defined shape factors. Compressibility analysis, angle of repose and shear test evaluate the relative flow properties of ground pine chip and ground pine pellet particles. The characterization results in this research express a functional relationship between physical properties and flowability of biomass particles. The output of this study is useful in modeling the thermochemical conversion of single particles. The flowability analysis helps to design and optimize operation of feeding systems in power plants.

2. Material and Methods

2.1. Sample Preparation

Pine wood chips (30x30x5 mm) and commercially produced pine wood pellet (diameter of 6 mm and lengths of 12-24 mm) were supplied by Fiber Co. industrial company (North Vancouver, BC, Canada). Upon their arrival at the lab, the pine chips were dried in a THELCO laboratory PRECISION oven (Thermo Electron Corporation, Model 6550) at 80 °C down to 4-5% moisture content. After cooling, the dried chips were crushed in a hammer mill (Glen mills Inc., USA; Model: 10HMBL) installed with four screen sizes with circular perforations of 3.2, 6.3, 12.7 and 25.4 mm. The moisture content of received pellets were 5% and thus were not subjected to oven drying. Wood pellets were ground using the same hammer mill with three grinder screen sizes of 3.2 mm, 6.3 mm and 12.7 mm. The perforation of 25.4 mm grinder screen size was too large for pellet grinding.

The hammer mill that used to grind the chips and pellets was connected to a data logging system that recorded the consuming grinding energy. 5 kg of material were fed to the grinder during 2 minutes time interval using a vibratory feeder. The consumed energy was averaged in

the steady-state period. The average value of steady-state consumed energy (kJ/s) was divided by feeding rate (kg/s) and the specific grinding energy, P (kJ/kg), was calculated.

2.2. Particle Size Distribution

The hammer mill produces mixed ground particles. To determine the particle size distribution of each mixed ground sample, a tap sieve shaker (Ro-Tap RX 94) installed with different mesh sieves of 0.25, 0.5, 1, 1.4, 2, 2.8 and 6.3 mm was used. The meshes were arranged from smallest to largest mesh size. The shaker subjected samples to oscillation and tapping for 10 minutes, following the procedure outlined in ASABE ANSI Standard S319.3. The mass retained on each sieve was weighed. The mean mass diameter of particles was calculated using equation (1) [16], where x_i is the retained mass fraction of particles in the ith interval of mean diameter (d_i) between two sieves.

$$d_{sv} = \frac{1}{\sum x_i / d_i} \tag{1}$$

2.3. Image Processing

A representative 500 g of mixed ground materials were sieved using a 0.25 mm sieve. Igathinathane et al. [19, 41] mentioned that very small particles (<0.25 mm) may agglomerate and does not result a very accurate and reliable image analysis. The retained particles were spread on a flat tray to a uniform depth. The samples were divided to 16 sub-parts. Using a spoon, a group of particles were picked randomly from the sample to be pictured. All ground pellet samples (ground using 3.2, 6.3 and 12.7 mm screens) and chip samples ground using 3.2 and 6.3 mm screens were observed and pictured by a high resolution microscope. The chip particles that were ground using 12.7 and 25.4 mm screens were too large to be pictures by microscope. The maximum vision range of microscope was limited to 13-15 mm. Therefore,

chip particles ground using 12.7 and 25.4 mm screens were pictured as a group of separated particles by a high resolution scanner connected to a computer. Pictures were analyzed using an image processing software (ImageJ ver. 1.49h, National Institutes of Health, USA). Figure 2 shows the full algorithm of image processing. ImageJ software was calibrated using a known-dimension kit to convert the pixel length to millimeter. Image processing requires the binary images. In binary pictures, zero is assigned to the background's pixels and one is assigned to the pixels inside the particle's boarder. Colored images are converted to grey-scale image by a "RGB Stack" command. Grey scale picture converted into binary image by proper thresholding. Igathinathane et al. [19] explained that using the same thresholding procedure may increase the halo effect around the particles and underestimate the particle's dimensions. Then, a specific thresholding procedure was examined and implemented for each particle picture.

The dimensions of the ground particles were measured using the ellipse fitting technique in ImageJ software. The ellipse fitting plugin fits the best ellipse on the particle's picture. ImageJ gives the major axis and minor axis of ellipse that are representative of particle length (L) particle width (W), respectively. Figure 3 shows the dimensions measured in image analysis of a single particle.

2.4. Shape Factors

Using the outputs of image analysis, aspect ratio (AR), W/d_{gs} and L/d_{gs}, sphericity (ψ), circularity and roundness are calculated. d_{gs} (mm) is the grinder screen opening size. The most common shape factor is aspect ratio (AR). AR of particles are calculated using equation (2).

$$AR = W/L \tag{2}$$

where W (mm) and L (mm) are width and length of particles, respectively. AR value ranges between zero and one. Equivalent diameter is the diameter of a sphere that has the same volume-to-surface area ratio as the particle of interest. Owing to difficulty in measuring the surface area and volume of irregular-shape particles, a two dimensional equivalent spherical diameter (d_{eq}) is suggested by Biagini et al. [2] in equation (3)

$$d_{eq} = (\mathrm{L} \times \mathrm{W}^2)^{1/3} \tag{3}$$

Sphericity (ψ) of a particle indicates how closely the particle's shape looks like a sphere. Sphericity is defined as surface area of a sphere having the same volume of the particle (S_v) divided by the actual surface area of the particle (S_s) [22, 42]. A quicker method is proposed based on the two dimensional measurements of inscribed circle diameter (d_{CI}) and circumscribed circle diameter (d_{CC}) [21]. Inscribed circle is the largest circle that fits inside the particle's boundary. Circumscribed circle is smallest circle that encloses the particle's boundary. This technique determines the sphericity as ratio of square root of inscribed (d_{CI}) and circumscribed (d_{CC}) circle's diameter [16].

$$\psi = \sqrt{d_{CI} / d_{CC}} \tag{4}$$

Circularity (ϕ) is another shape factor that indicates angularity or sharpness of corners in a two dimensional analysis.

$$\phi = 4\pi A \big/ P^2 \tag{5}$$

where A and P are the projection area and perimeter of the particle, respectively. ImageJ counts the number of pixels inside the particle's boundary to determine area (A) and counts the number of pixels of particle's boundary to determine perimeter (P). Circularity is 1 for a sphere

and 0.785 for a square. Similar to circularity, Roundness is an indicator of angularity of a particle profile and shows the presence of sharp corners. The calculation of roundness based on individual curvature in particle profile is a time consuming and laborious work. There is a correlation in literature, equation (6), to determine the roundness in an easier way.

$$Roundness = 4A \big/ \pi L^2 \tag{6}$$

Similar to circularity, the largest value for roundness is 1 for a circular object. It is important to mention that comparison of the same shape factor at different studies could be very confusing. Different concepts are assigned to the same shape factor. For example, Biagini et al. [2] defined the roundness as (1/circularity) of what has been defined in this study. It is important to know what concept is addressed for each shape factor.

2.5. Pycnometer Density

The solid density of a particle refers to the mass of the particle divided by the volume of the particle's skeleton ($\rho_s = m_p / V_s$). In a gas pycnometer, the helium or nitrogen gases penetrate into mesopores (2-50 nm), but their access to micropores (<2 nm) and closed pores are limited. Therefore, the pycnometer density is a value between particle density and solid density. Pycnometer density is particle's mass divided by summation of solid volume and volume of closed pores, $\rho_p = m_p / (\mathrm{V}_s + \mathrm{V}_{cp})$ [16]. Pycnometer density was measured using a nitrogen gas pycnometer, model MVP-D160-E (Quantachrome Instruments, Boyton Beach, FL, USA). The unit had two chambers. First was reference cell with volume of 11.67 cm^3. Second was sample cell with volume of 29.42 cm^3. The pressure of reference cell was adjusted to around 17 psig. The pressurized nitrogen diffused into the bed and pores of particles in the sample cell. Then, the summation volume of skeleton and closed pores was calculated using the pressure difference

measurement and ideal gas law. All of density analysis were done in five times to show the reproducibility of results.

2.6. Bulk Densities

Loose bulk density refers to the mass of bulk of sample divided by its aerated volume ($\rho_{lb} = m_b / V_{lb}$). Constant mass of 200 g (m_b) of each ground sample was smoothly poured in a graduated plastic vessel. The volume of loosed sample was recorded (V_{lb}). Then, the vessel was dropped under its own mass from the height of 10 cm at different times. The volume of tapped particles was recorded in each 10-tap interval (V_{tb}). The vessel was tapped 100 times. The tapped bulk density was obtained by dividing mass of sample over volume of tapped sample ($\rho_{tb} = m_b / V_{tb}$). The Hausner ratio (HR) and Carr compressibility index (CCI) were calculated using equations (7) and (8).

$$HR = \frac{\rho_{tb}}{\rho_{lb}} \tag{7}$$

$$CCI = (1 - \frac{\rho_{lb}}{\rho_{tb}}) \times 100 \tag{8}$$

where ρ_{lb} (g/cm^3) and ρ_{tb} (g/cm^3) are loose and tapped bulk density of ground biomass particles.

2.7. Angle of Repose

In this study, an equipment was developed to conduct the angle of repose analysis of ground biomass particles. A 50 g representative sample of each ground chip and ground pellet particles to be tested was weighed out. The ground materials were poured gently on an inclined converging sheet that was vibrating. The sheet was installed with angle of 45°. The particles were directed to a funnel and then dropped by gravity force on a graduated plate. A semi-cone

pile was made. Due to the scattered shape of pile edge, reading the height and radius of pile was challenging. To solve the problem, the formed piles were pictured and the angle of repose was measured using ImageJ software.

3. Results

3.1. Particle Size Distribution

Figure 4 shows the particle size distribution (PSD) of ground chip and pellet particles obtained from mechanical sieving. The PSD of chip particles is wide spread, since the PSD of pellet particle is narrow spread. Pellet particles were less sensitive to the grinder screen size and had a homogenous PSD, since the chip particles spread in a very wide size range. It must have been due to the fact that the pellets' internal particles were ground before pelleting process using a 3-5 mm grinder screen [16, 43]. In general, ground pellet particles are smaller and have a narrower size distribution than chip particles. Mass-averaged diameter of ground chip particles were 0.58, 0.87, 1.59 and 2.29 mm for grinder screens of 3.2, 6.3, 12.7 and 25.4 mm, respectively. The mass-averaged diameter of ground pellet particles were 0.50, 0.56 and 0.64 mm for grinder screens of 3.2, 6.3 and 12.7 mm, respectively.

It looks that the 3.2 mm grinder screen produced the chip and pellet particles with similar PSD. For both, most of particles retained on sieve sizes of 0.25 mm and 0.5 mm. About 36-39% of chips and pellet particles ground with 3.2 mm grinder screen were passed through the 0.5 mm sieve. Table 1 also lists the fraction of particles that passed through the sieves of 1 mm and 2 mm. Several researchers recommended to reduce the size of biomass particles to smaller than 2 mm in order to minimize the biochar yield during pyrolysis/combustion [7-12]. The PSD output shows that the pellet particles always have >88% particles smaller than 2 mm. But, grinding wood chip with only grinder screens of smaller than 6 mm gives an appropriate PSD for thermal

conversion. Although, the sieving does not give the actual size and shape of particles necessarily and a more accurate size analysis technique is required to measure the real size of particles and characterize the shape of ground particles.

3.2. Grinding Energy

The pine chips and pine pellets were ground using a hammer mill installed with grinder screen sizes of 3.2, 6.3, 12.7 and 25.4 mm. Figure 5 shows the specific grinding energy (kJ/kg) to grind the pine chips and pine pellets with different grinder screens. Grinding of pine chips took significantly more energy than pellets. For example, grinding chips with 3.2 mm screen consumed about 7 times more energy than grinding pellets with the same grinder screen. The grinding energy of chips declined drastically from 124.1 kJ/kg for 3.2 mm grinder screen to 28.2 kJ/kg for 25.4 grinder screen. The grinding energy of pellets declined very smoothly from 16.7 kJ/kg for 3.2 mm grinder screen to 12.2 kJ/kg for 12.7 grinder screen. Two empirical equations were fitted with experimental grinding energy data. The pellets are made from the previously-crushed particles and were easier to grind. Grinding is an energy-taking step in preparing feedstock for pyrolysis or combustion units. The lower grinding energy of pellet significantly affects the decision regarding of using ground chips or ground pellet in a pyrolysis process.

3.3. Pycnometer Density

Table 3 lists the pycnometer density values of ground chip and pellet particles. The pycnometer density increased with grinding level. Size reduction broke the particles' structure that opened the close pores and increased the number of accessible pores for the pycnometer gas to penetrate. The density of ground chip particles reduced from 1.329 20 1.051 g/cm^3 when the grinder screen changed from 3.2 mm to 25.4 mm. The pycnometer density of ground pellet

particles also dropped slightly with grinding level. The density of ground pellet particles reduced from 1.429 to 1.404 g/cm^3 when the grinder screen changed from 3.2 mm to 12.7 mm.

There were two differences between pycnometer densities of chip and pellet particles. First, the grinding affected the density of chip particles more than pellet particles. It came from the fact that the internal particles of pellet were ground before pelleting and further grinding of pellets did not change the pycnometer density significantly. Second difference is the higher pycnometer density of pellet particles than chip particles. The second difference was resulted from the structural transformation in pelleting process. The applied force in the densification process smashed the cellular structure of fibers and reduced the pore volume of particle. Figure 6 shows the micrograph of single chip and pellet particle.

3.4. Particle Dimensions and Shape

The output of single particle shape analysis showed that the actual dimensions and shape of ground chip and pellet particles were not similar. The differences were visually observed and showed in Figure 3 and Figure 7. Ground chip particles had mostly rectangular shape. The low values of sphericity, circularity and roundness shape factors in Table 1 showed that the ground chip particles were far from spherical shape. Sphericity and roundness of all ground chip particle were in range of 0.44-0.45 and 0.21-0.25, respectively. The particles had a small aspect ratios of 0.21-0.22. Chip particles had a uniform width alongside their length. Ground pellet particles were completely irregular in shape. Pellet particles look more spherical. Sphericity and roundness of pellet particle were in range of 0.70-0.73 and 0.62-0.70, respectively. Sphericity and roundness of pellet particles were about 1.5-2 and 3-3.5 times of chip particles, respectively. The average aspect ratio values of pellet particles was 0.6-0.64 that was about 3 times of chip

particles. Due to the curvature and surface roughness of pellet particles, they had different width alongside the particle's length.

Ground chip and ground pellet particles were also different regarding their dimensions. Looking at the W/d_{gs} and L/d_{gs} shape factors in Table 1 shows that all dimensions of the pellet particles were smaller than the grinder screen opening size. While, the length of chip particles were always larger than grinder screen opening size. Figure 8 shows that width (W) and length (L) of both chip and pellet particles increased linearly with grinder screen opening size (d_{gs}). The empirical correlations have been developed to express the average width and length of ground particles as a function of grinder screen size, presenting in Table 2.

Even though the mechanical sieving separates the particles based on their width, Figure 8 shows that the d_{sv} obtained from mechanical sieving was always less than the actual width of particles. It seems that mechanical sieving underestimated the actual width of particles. The d_{sv} was about 1/2 of particle width for chip particles and 1/3 of width for pellet particles. Unlikely, the d_{eq} was always a value between width and length. Four linear experimental correlations are presented in Table 2 to show the variation of average width (W), average length (L), mass-averaged diameter (d_{sv}) and equivalent spherical diameter (d_{eq}) with grinder screen size (d_{gs}).

3.5. Flow Characterization

Table 3 lists the loose and tapped bulk densities of chip and pellet particles. The loose bulk density of both chip and pellet particles increased with increasing particle size. The bulk density of pellet particles are higher than chip particles. Two reasons explain higher bulk density of pellet particles. First, single pellet particles are denser than chip particles. Second, chip particles have a much lower aspect ratio (W/L) and their random settlement in a bulk makes a more

porous bulk. During the tapping process, the bulk density of particles increased. Smaller particles took more time and tapping to reach to their final compaction. Chip particles ground using 3.2 and 6.3 mm grinder screens continued to compress till about 70-80 taps, while larger chip particles reached to their stable compaction after about 20 taps. All ground pellet samples reached to their final compression after 30-40 taps. The final tapped bulk density after 100 taps was used to calculate the HR and CCI.

Figure 9 displays that HR and CCI decreased with particle size. Larger particles had lower tendency to make a compact bulk. Smaller particles were able to fill very small pores in the bulk and increase the density. The same trend was also observed for pellet particles. Although, the variation in pellet particles' compressibility was less than chip particles. It must have been due to the very similar particle size distribution of pellet particle ground with different grinder screens.

Particle size was not the only affecting factor on the compressibility. Chip and pellet particles ground with 3.2 mm grinder had a similar PSD; but, chip particles compacted significantly more than pellet particles. Chip and pellet particles ground with 3.2 mm grinder screen compacted 50% and 26%, respectively. It seems that the level of compression highly depended on shape of the particles. Pellet particles were more spherical and rather than more-needle shape of chip particles. It seems that higher length promoted the compression of particles. The effect of particle shape was not observed just for particles with same size and particle size distribution. Comparing the particle size, particle shape and compaction of chip and pellet particles ground with 6.3 and 12.7 mm grinder screens confirmed that particle shape was more important than particle size in tapping compressibility. Grinder screens of 6.3 and 12.7 mm made pellet particles much smaller than chip particles. But, still the pellet particles compacted less and had a significantly smaller HR and CCI values.

The biomass particles are cohesive and have an interaction among each other. Angle of repose indicates the relative cohesiveness of the particles under a normal force [27]. Table 3 lists the AOR of chip and pellet particles. AOR of chip particles decreased from 61.00° to 45.89° when grinder screen changed from 3.2 mm to 25.4 mm. It is in agreement with literature that reported increase in particle size is accompanied by decrease in cohesiveness and then AOR [16, 30]. Small particles have higher specific surface area that boosts the contact and cohesiveness among the particles. Due to very similar PSD of pellet particles, the AOR values of pellet particles were practically independent of grinder screen size. Similar to the compressibility, the effect of particle shape was more significant than particle size. Chip and pellet particles ground with 3.2 mm grinder screen had AOR values of 61° and 48°, respectively. This difference between AOR of chip and pellet particles with the same PSD sourced from their different particle shape. The same amount of AOR reduction was observed when grinder screen switched hugely from 3.2 mm to 25.4 mm for chip grinding.

4. Conclusions

In this study, the size, shape and flow properties of ground pine chip particles and ground pine pellet particles were characterized. To produce the same size of particles, the wood chips consumed about 7 times more energy to be ground than wood pellets. Ground pellet particles were smaller and had a narrower particle size distribution than ground chip particles. More than 88% of all ground pellet samples passed through a 2 mm sieve. Wood chips need to be ground with grinder screens of smaller than 6 mm to produce an appropriate PSD for thermal conversion.

Pelletization reduced the variability in size distribution, particle shape and flow properties of ground particles. Pellet particles are more circular shape and have the aspect ratios (W/L) of

about 3 times of long and narrow chip particles. Needle-shape chip particles cause the bulk to compress more. Angle of repose tests showed that pellet particles are less cohesive and flow easier than chip particles. Small ground chip particles may cause flow issues in a bulk of particles. Flow characterization showed that effect of particle shape on flow properties of particles is more than the particle size.

Acknowledgement

This study has been conducted under financial support of Biofuel Network Canada (BFN) and natural Sciences and Research Council of Canada. The authors also gratefully acknowledge contribution of Fibreco Inc. for providing the wood chips and pellet samples.

Nomenclature

A	Area of projected particle, mm^2
AR	Aspect ratio of particle
d_{CI}	Inscribed circle diameter, mm
d_{CC}	Circumscribed circle diameter, mm
d_{eq}	Equivalent spherical diameter, mm
d_i	Mean diameter between two sieves, mm
d_{sv}	Mass-averaged particle size, mm
d_{gs}	Grinder screen size, mm
L	Particle length, mm
m_b	Mass of bulk of particles, g
m_p	Mass of particle, g
P	Perimeter of projected particle, mm
S_s	Surface area of a sphere having the same volume as the particle, m^2
S_v	Actual surface area of particle, m^2
V_{lb}	Volume of loosed bulk of particles, cm^3
V_{tb}	Volume of tapped bulk of particles, cm^3
V_p	Total volume of particle, cm^3
V_s	Volume of solid part of particle, cm^3
V_{cp}	Volume of closed pores of particle, cm^3
V_{op}	Volume of open pores of particle, cm^3
W	Particle width, mm
x_i	Retained mass fraction of particles in the ith sieve interval

Greek symbols

ϕ	Circularity
ψ	Sphericity
ρ_p	Pycnometer density, g/cm^3
ρ_{lb}	Bulk density, g/cm^3
ρ_{tb}	Tapped bulk density, g/cm^3
ρ_s	Solid density, g/cm^3

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Tables

Table 1 – Size and shape properties of ground chip and ground pellet particles using mechanical sieving and image analysis (mean ± standard deviation).

Parameters	Ground chip particles				Ground pellet particles		
d_{gs} (mm)	3.2	6.3	12.7	25.4	3.2	6.3	12.7
Mechanical sieving							
f_{1mm}(%) [1]	81.952	45.067	16.514	7.429	85.376	76.333	63.109
f_{2mm}(%) [2]	99.309	87.529	44.582	22.088	99.931	97.445	88.089
d_{sv} (mm)	0.580 ± 0.084	0.870 ± 0.102	1.590 ± 0.098	2.290 ± 0.123	0.500 ± 0.073	0.560 ± 0.062	0.640 ± 0.067
Image analysis							
W (mm)	1.031 ± 0.299	1.790 ± 0.542	3.108 ± 1.128	5.102 ± 2.502	1.536 ± 0.285	1.806 ± 0.612	2.415 ± 1.146
L (mm)	4.951 ± 1.464	8.278 ± 2.039	15.294 ± 4.741	25.555 ± 7.074	2.506 ± 0.569	3.115 ± 0.956	3.682 ± 1.031
AR	0.223 ± 0.090	0.222 ± 0.074	0.217 ± 0.091	0.211 ± 0.113	0.640 ± 0.165	0.622 ± 0.175	0.639 ± 0.185
d_{eq} (mm)	1.713 ± 0.407	2.949 ± 0.724	5.194 ± 1.453	8.483 ± 2.940	1.793 ± 0.262	2.139 ± 0.601	2.750 ± 1.113
W/d_{gs}	0.322 ± 0.093	0.284 ± 0.086	0.245 ± 0.089	0.201 ± 0.099	0.480 ± 0.089	0.287 ± 0.097	0.190 ± 0.090
L/d_{gs}	1.547 ± 0.457	1.314 ± 0.324	1.204 ± 0.373	1.006 ± 0.279	0.783 ± 0.178	0.495 ± 0.152	0.290 ± 0.081
Sphericity	0.443 ± 0.080	0.444 ± 0.075	0.457 ± 0.092	0.445 ± 0.115	0.729 ± 0.091	0.698 ± 0.083	0.717 ± 0.108
Circularity	0.287 ± 0.081	0.246 ± 0.069	0.336 ± 0.100	0.325 ± 0.114	0.400 ± 0.138	0.387 ± 0.106	0.320 ± 0.077
Roundness	0.253 ± 0.098	0.262 ± 0.089	0.217 ± 0.091	0.211 ± 0.113	0.706 ± 0.208	0.624 ± 0.179	0.683 ± 0.202

[1] f_{1mm}: Fraction of particles passed through the 1 mm sieve
[2] f_{2mm}: Fraction of particles passed through the 2 mm sieve

Table 2 – Empirical correlations of width (W), length (L), mass-averaged diameter (d_{sv}) and equivalent spherical diameter (d_{eq}) of ground chip and ground pellet particles as a function of grinder screen size (d_{gs}).

Dimension (mm)	Ground chip particles		Ground pellet particles	
	Correlation	R^2	Correlation	R^2
Width (W)	$W = 0.181d_{gs} + 0.608$	0.98	$W = 0.093d_{gs} + 1.231$	0.99
Length (L)	$L = 0.924d_{gs} + 2.528$	0.99	$L = 0.118d_{gs} + 2.224$	0.95
Mass-averaged diameter (d_{sv})	$d_{sv} = 0.077d_{gs} + 0.418$	0.95	$d_{sv} = 0.014d_{gs} + 0.460$	0.97
Equivalent spherical diameter (d_{eq})	$d_{eq} = 0.301d_{gs} + 1.003$	0.98	$d_{eq} = 0.100d_{gs} + 1.488$	0.99

Table 3 – Flow properties of ground chip and ground pellet particles (mean $\pm$ standard error).

Material	d_{gs} (mm)	Densities			Compressibility properties		Flow class (HR/CCI)	AOR (degree)
		Pycnometer (g/cm^3)	Loose bulk (g/cm^3)	Tapped bulk (g/cm^3)	HR	CCI (%)		
Ground chip particles	3.2	1.329 ± 0.007	0.138 ± 0.002	0.217 ± 0.002	1.572 ± 0.011	34.389 ± 0.453	Poor/Poor	61.00 ± 1.09
	6.3	1.322 ± 0.011	0.144 ± 0.001	0.209 ± 0.002	1.455 ± 0.007	31.281 ± 0.352	Poor/Poor	56.97 ± 0.63
	12.7	1.265 ± 0.006	0.145 ± 0.001	0.188 ± 0.003	1.293 ± 0.022	22.667 ± 1.333	Good/Fair	52.54 ± 0.75
	25.4	1.051 ± 0.011	0.157 ± 0.002	0.188 ± 0.001	1.196 ± 0.039	16.349 ± 2.698	Excellent/Good	45.89 ± 1.42
Ground pellet particles	3.2	1.429 ± 0.008	0.390 ± 0.002	0.493 ± 0.007	1.266 ± 0.032	20.984 ± 1.967	Good/Fair	48.02 ± 0.67
	6.3	1.427 ± 0.015	0.405 ± 0.002	0.493 ± 0.001	1.219 ± 0.018	17.976 ± 0.942	Excellent/Fair	47.99 ± 0.31
	12.7	1.404 ± 0.007	0.422 ± 0.005	0.503 ± 0.007	1.194 ± 0.018	16.203 ± 1.258	Excellent/Good	47.57 ± 0.26

Figures:

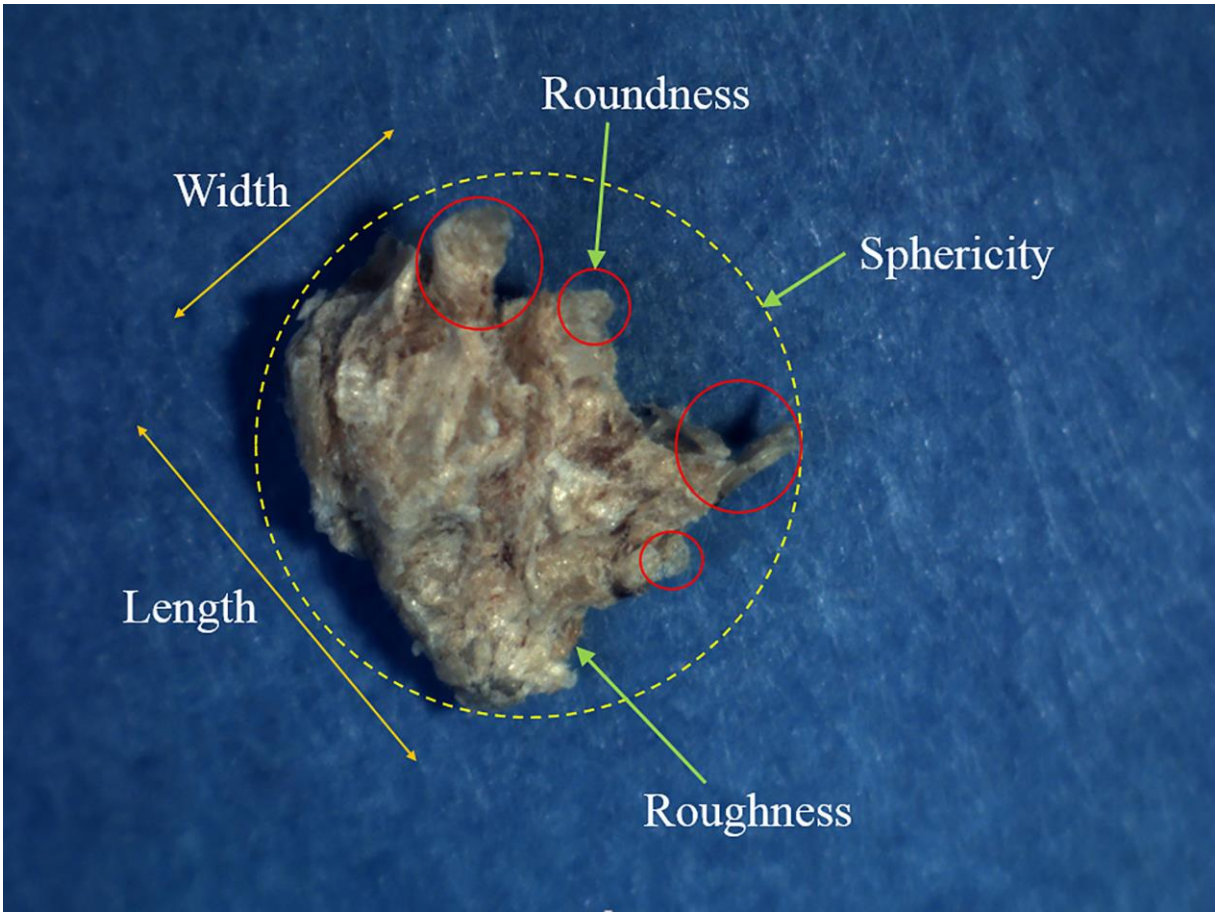


Figure 1 – Particle shape characterization in terms of dimensional ratios, sphericity, roundness and roughness.

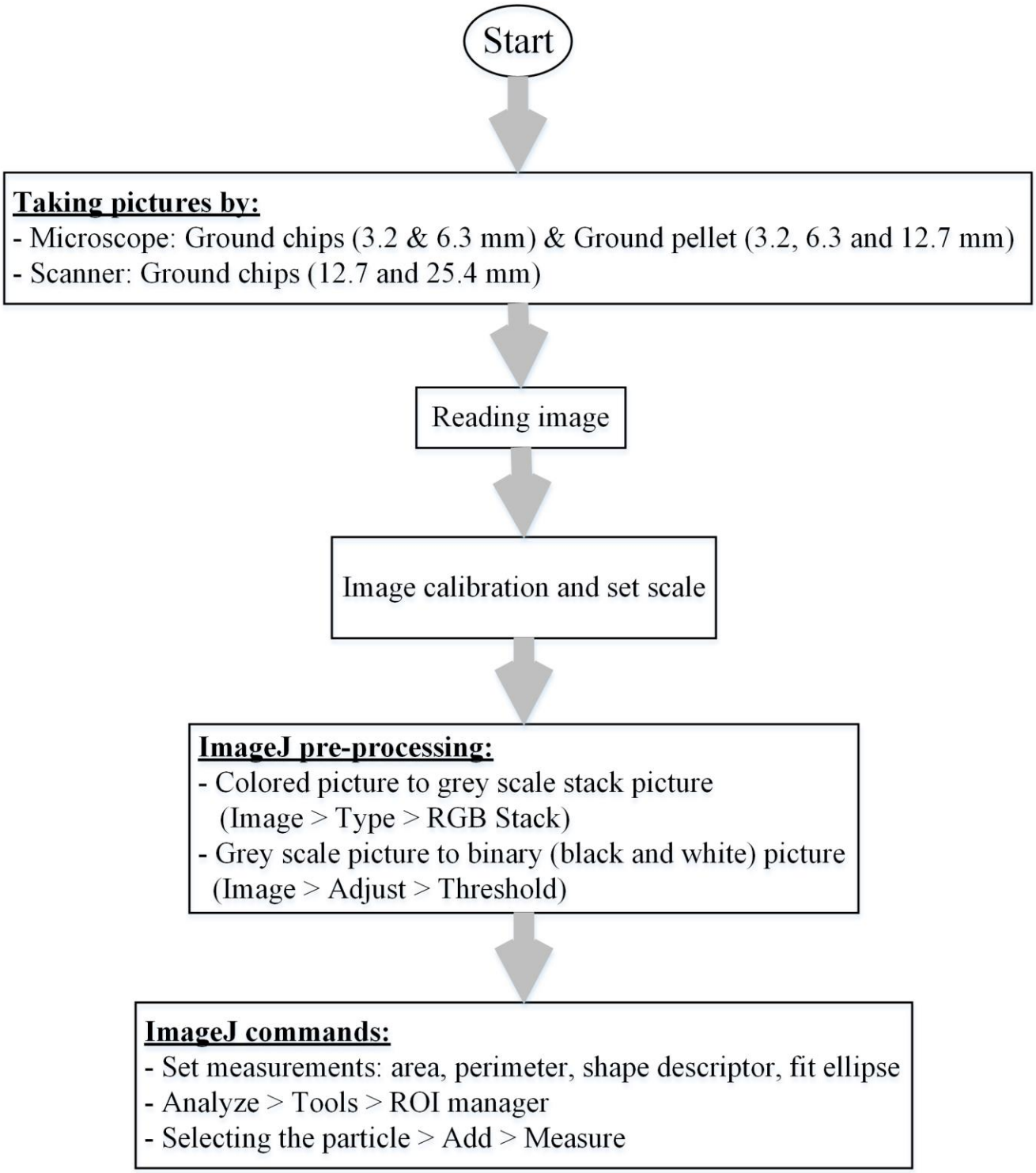


Figure 2 – Procedure flowchart of particle image analysis using ImageJ software.

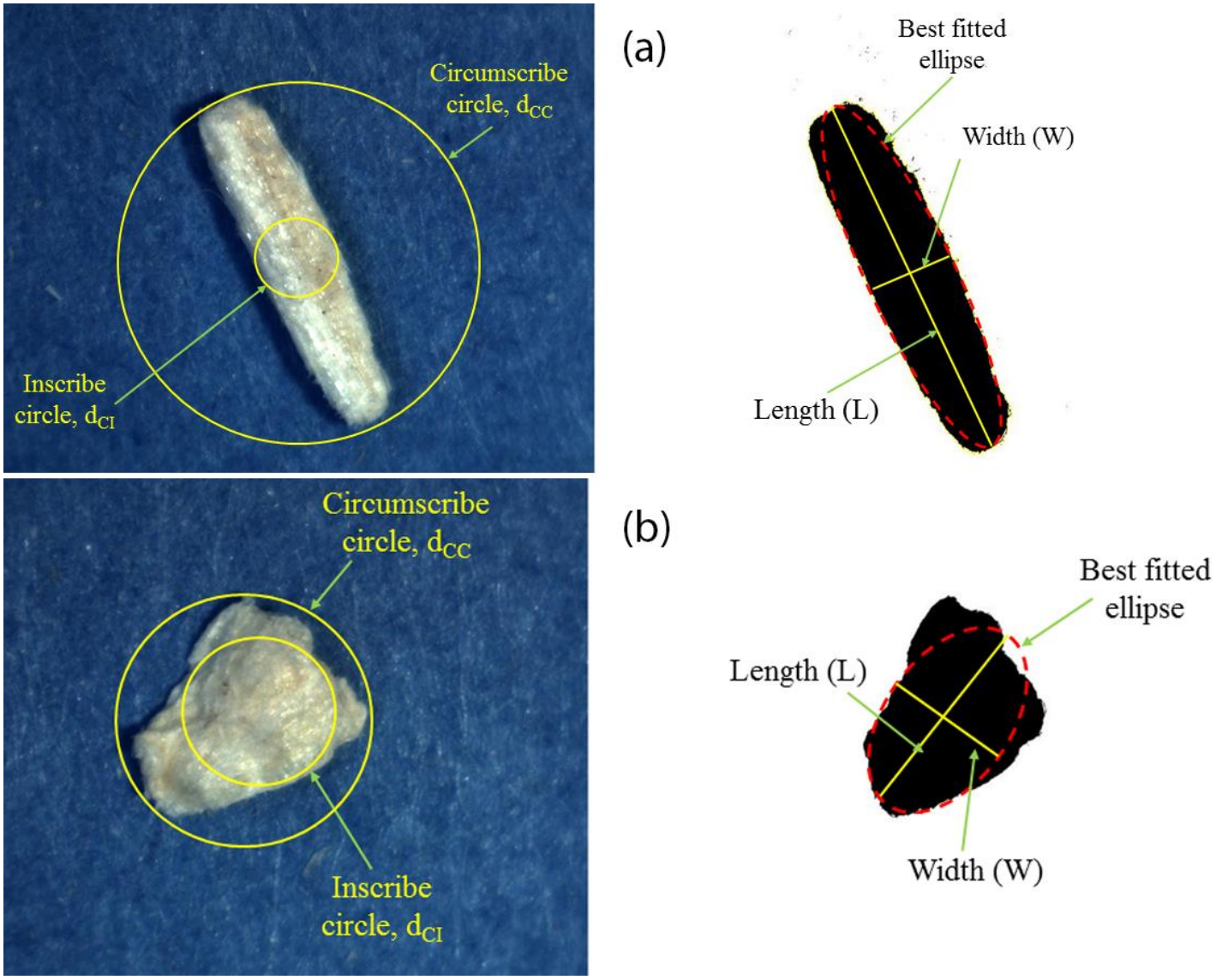


Figure 3 – Determining width (W), length (L), inscribed circle (d_{CI}) and circumscribed (d_{CC}) circle in a (a) ground chip particle and (b) ground pellet particle.

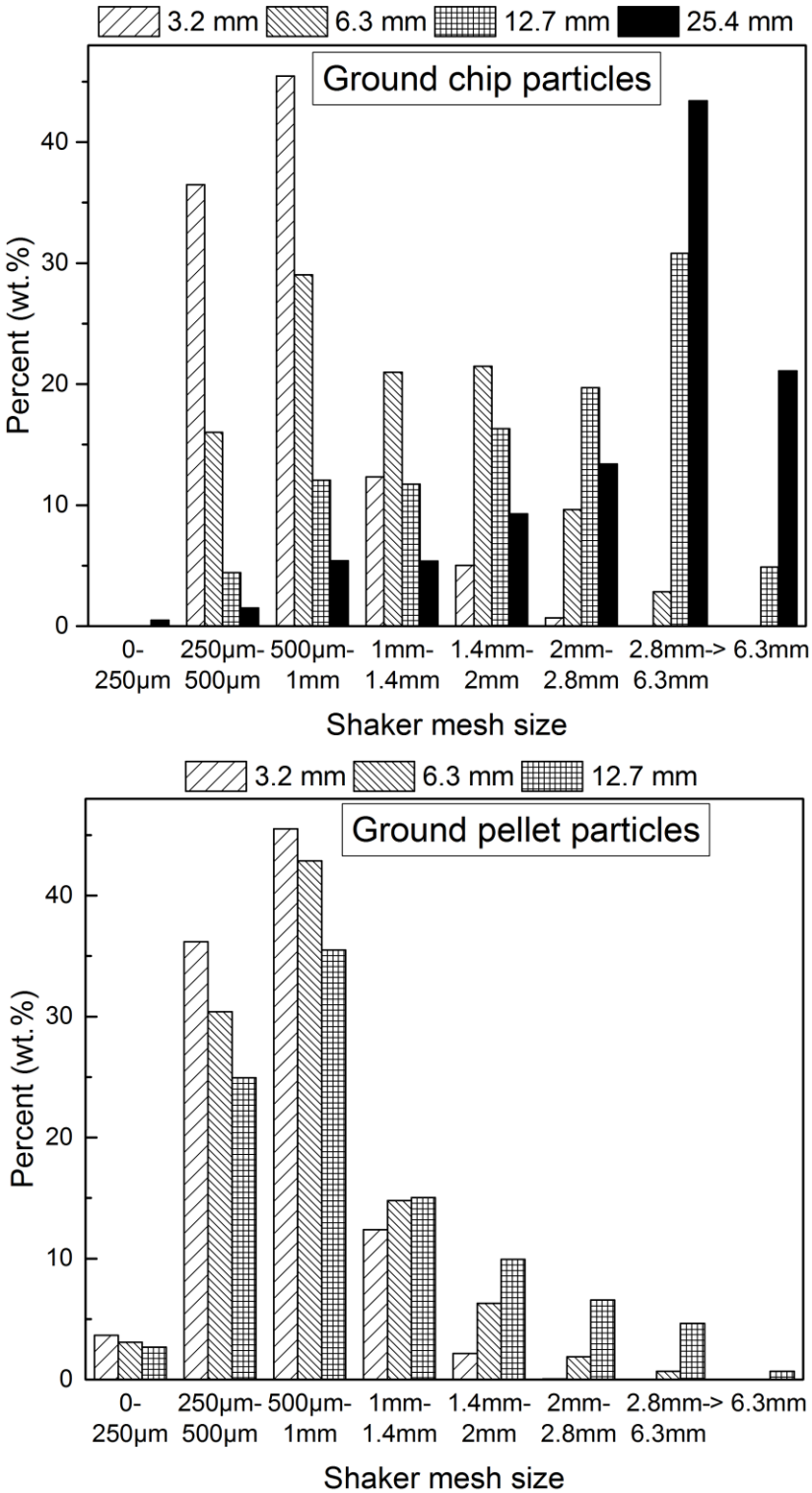


Figure 4 – Particle size distribution (PSD) of ground chip and pellet particles obtained by mechanical sieving.

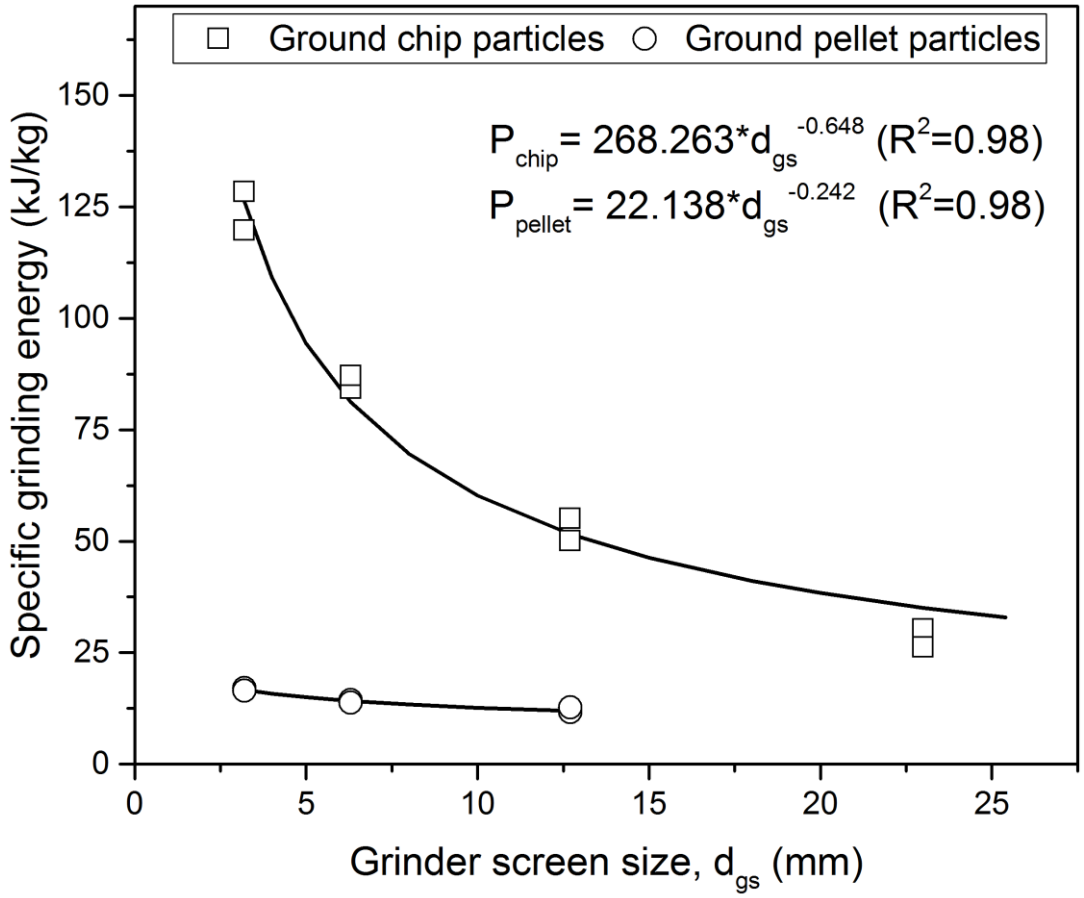
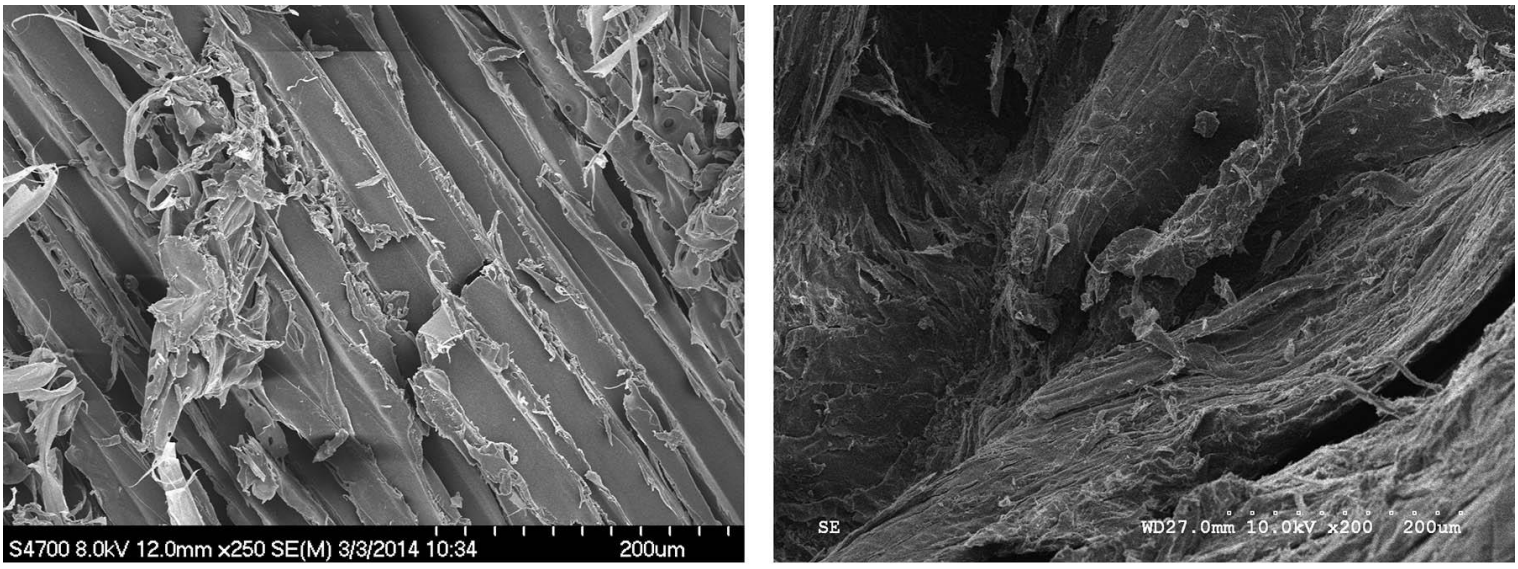
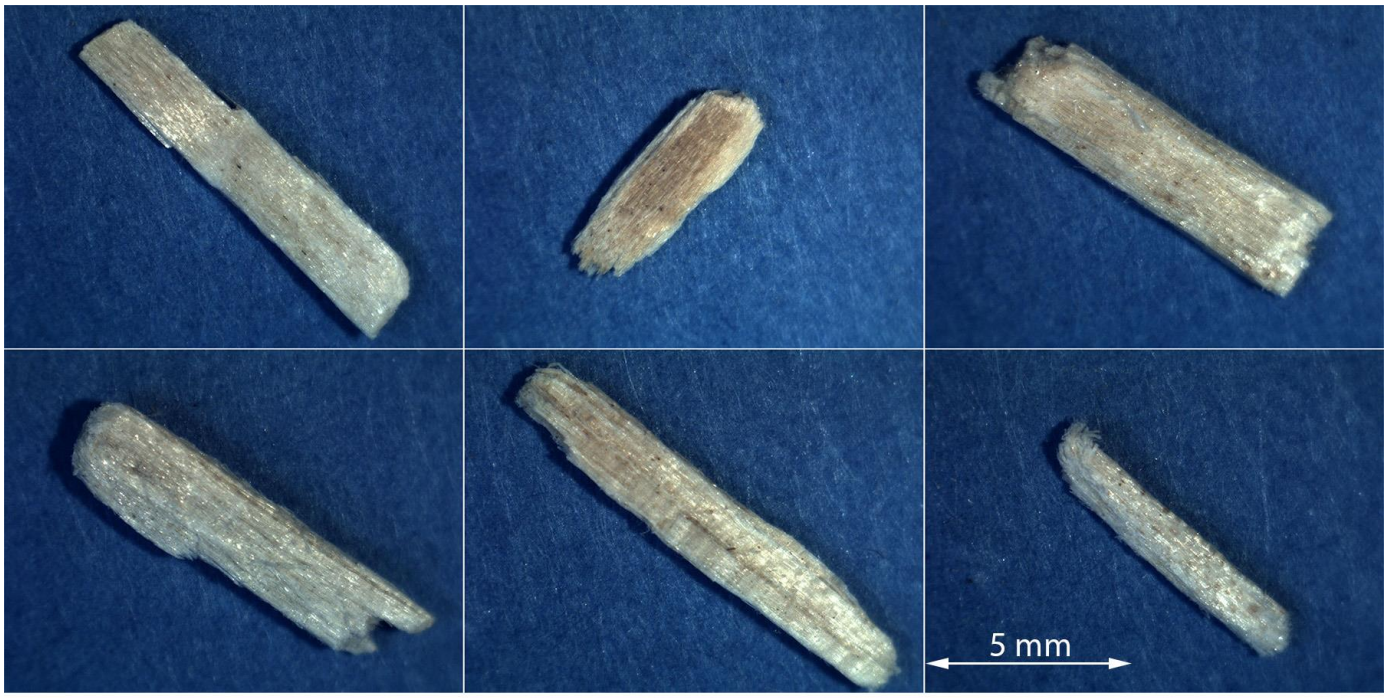


Figure 5 – Specific energy requirement to grind the chips and pellets using the grinder screen sizes of 3.2, 6.3, 12.7 and 25.4 mm.



Ground chip particle Ground pellet particle

Figure 6 – SEM picture of ground pine chip and pellet particles.



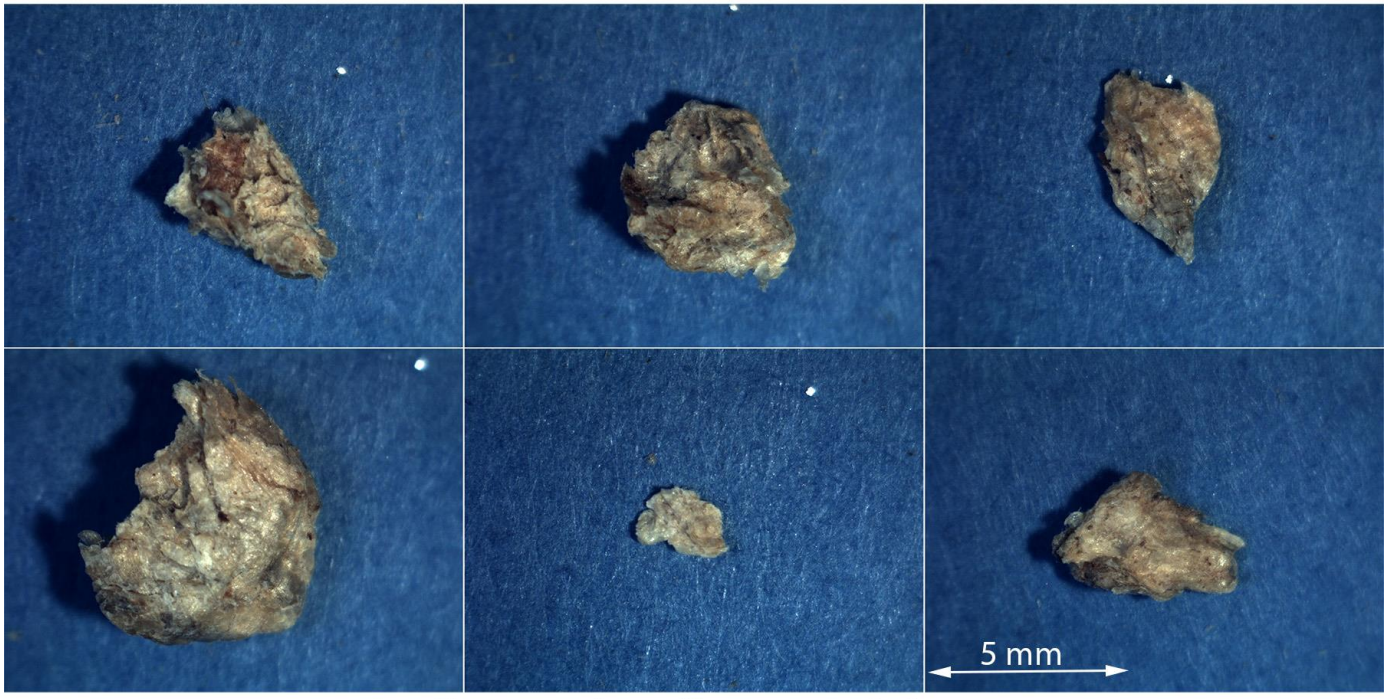


Figure 7 – Microscopic pictures of ground pine chip and pellet particles.

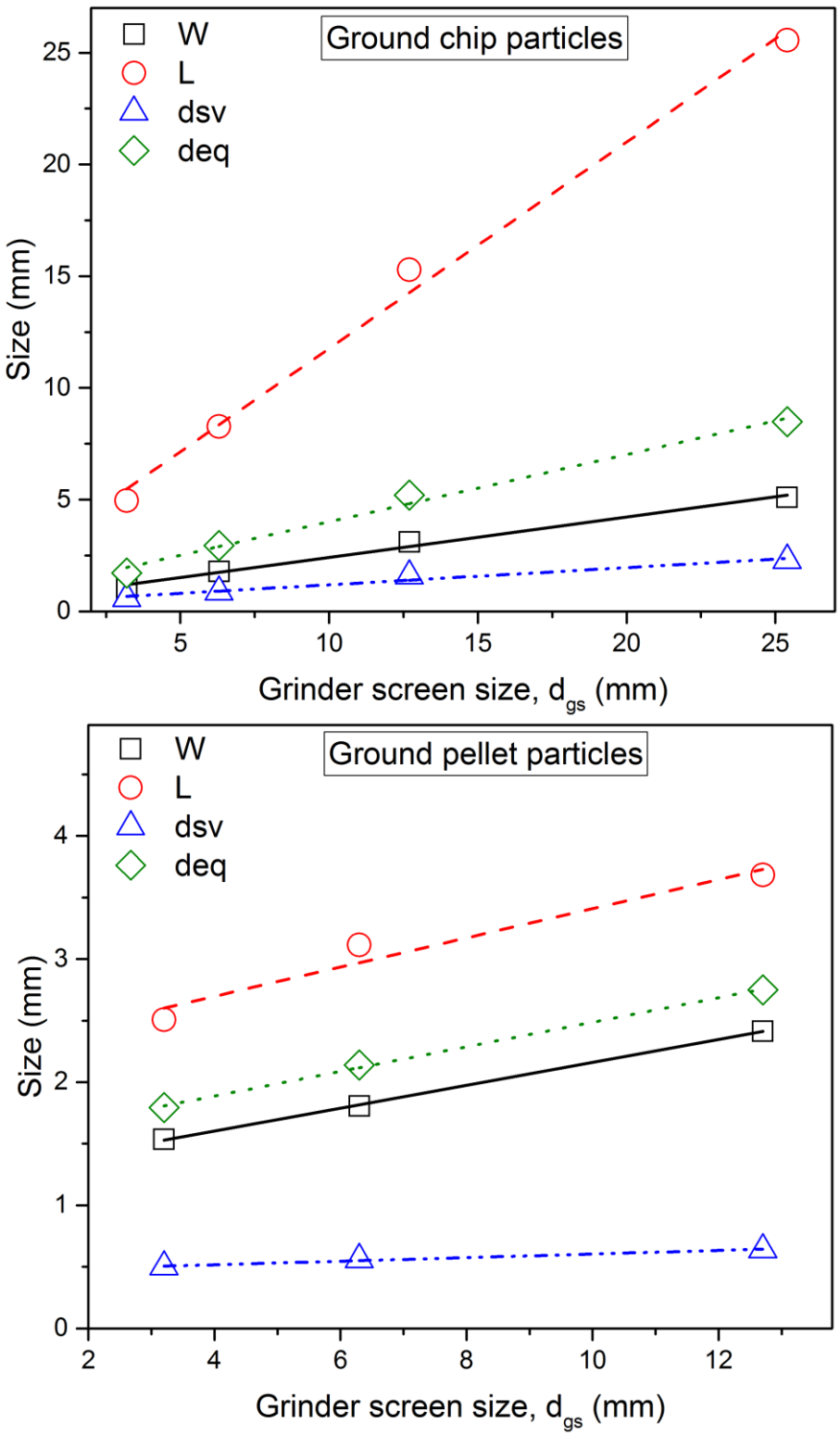


Figure 8 – Variation of average width (W), average length (L), mass-averaged diameter (d_{sv}) and equivalent spherical diameter (d_{eq}) with grinder screen size (d_{gs}) for ground chip and pellet particles.

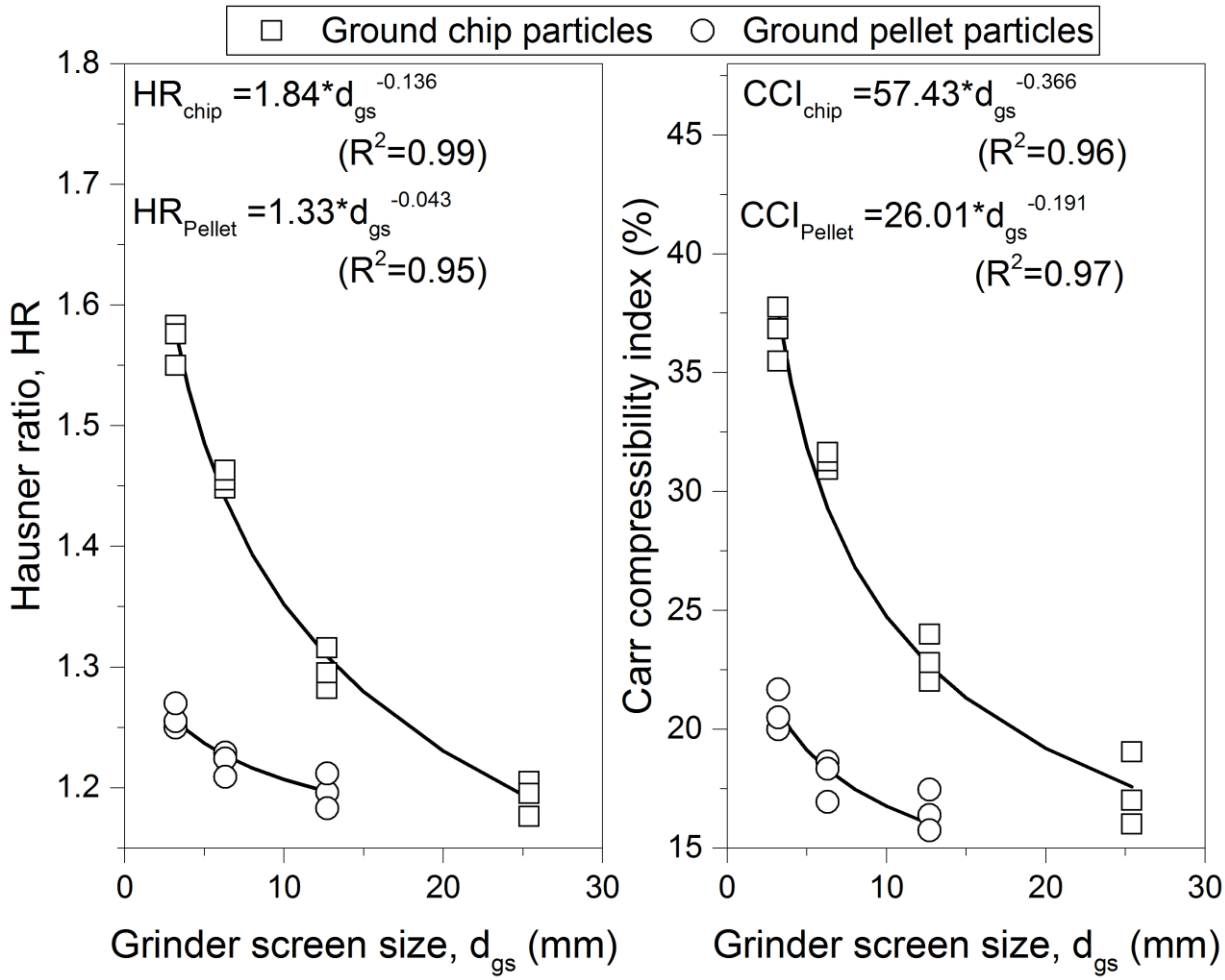


Figure 9 – Hausner ratio (HR) and Carr compressibility index (CCI) of ground chip and pellet particles.

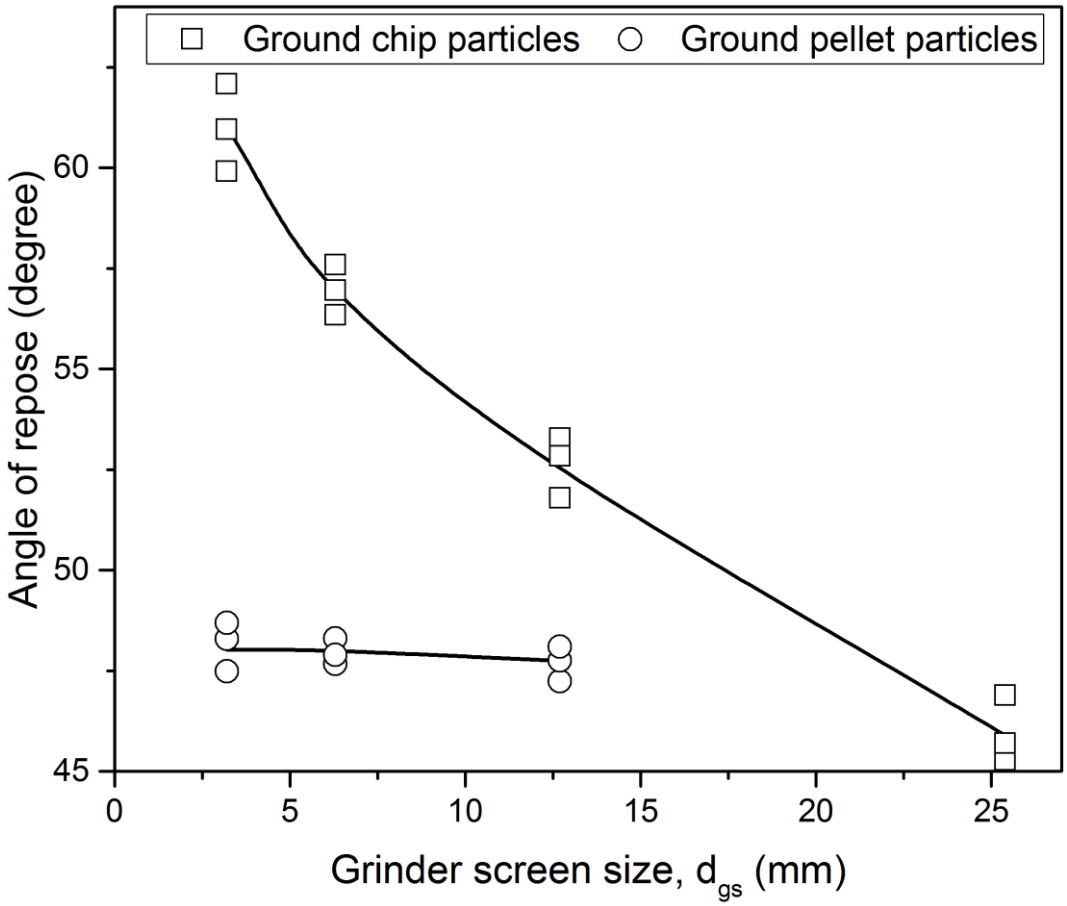


Figure 10 – Angle of repose (AOR) of ground chip and pellet particles.